Suparno Ghosh

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ABOUT ME

I am a third-year undergraduate at IISER Kolkata, majoring in Mathematics and minoring in Computer Science. I am specifically interested in Structural Graph Theory and it's applications in non-mathematical disciplines. I am fascinated at how Graph theory interplays unimaginably with every other branch of Mathematics as well. Beyond my usual coursework and projects, I enjoy Philosophy, Bengali Literature, playing chess and directing and acting in theatres.

EDUCATION AND TRAINING

16/08/2023 – CURRENT Kolkata, India

BS-MS Indian Institute of Science Education and Research, Kolkata

- Majoring in *Mathematics (Pure)*
- Minoring in *Computational and Data Science*
- Pre-Majored in *Physics*

Website www.iiserkol.ac.in | **Field of study** Mathematics | **Final grade** Current SGPA: 9.80. Current CGPA: 9.03 |

Level in EQF EQF level 6

26/05/2025 – CURRENT Bangalore, India

STUDENT INTERN Tata Institute of Fundamental Research - Centre for Applicable Mathematics

- Studied *Calculus of Variations* using techniques derived from *Advanced Several Variable Calculus* and *Differential Geometry*

Level in EQF EQF level 4

SKILLS

LEAN | Python (Advanced) | C++ | JavaScript | LaTeX (Advanced) | Discrete Mathematics | Graph Theory (Advanced) | Structural Graph Theory (Undergraduate_ | Digraph Theory | MATLAB | SageMath | Semi-Definite Programing (SDP) | LP and Convex Optimisation | Linear Algebra (Advanced) | probability theory | Statistics (Intermediate) | Functional Analysis | Algorithms and Data Structures

LANGUAGE SKILLS

Mother tongue(s): **BENGALI**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C2	C2	C2
FRENCH	A2	B1	A2	A2	B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

CERTIFICATIONS

Indian Institute of Science, Bangalore , 25/05/2025

Summer School on Approximation Algorithms

Selected for an enriching summer school on Approximation Algorithms. Fascinated by the use of graph theoretic analysis in NP-hard problem solving. Got to learn about various techniques of approximations on graph and digraph problems - LP relaxation techniques and Primal-Dual Algorithms, applied SDP, Goemans-Williamson Algorithm, FPT Approximations, etc. Also got introduced to Multicut, Local Search, Online Algorithms and Cheeger's Inequality.

Mode of learning: Presential

Links https://media.licdn.com/dms/image/v2/D4D2DAQEttkj9GWwpug/profile-treasury-image-shrink_800_800/B4DZncsN.6JEAY-/0/1760344208053?e=1766397600&v=beta&t=sEzHp9p-JQll1Y--QLI5qNC35sfWHsmYszzlSVgKXKw | https://algo.csa.iisc.ac.in/summerschool_25/

Indian Academy of Sciences, Bengaluru, 2025

Summer Research Fellow at Indian Academy of Science

Mode of learning: Project based

Link https://media.licdn.com/dms/image/v2/D4D2DAQFUi08F2jcuHQ/profile-treasury-image-shrink_1280_1280/B4DZnctJnRKoAQ-/0/1760344455024?e=1766397600&v=beta&t=gKvujdXsQ4WhmdN6nEYyQwk4IYDbcpM_8SHRYQNRdhM

HONOURS AND AWARDS

2023

INSPIRE Scholarship – Department of Science and Technology, Government of India

2025

Summer Research Fellowship 2025 – Indian Academy of Science, Indian National Science Academy and The National Academy of Sciences

Selected for Summer Research in the field of Mathematics to work at TIFR-CAM, Bengaluru.

Link https://webjapps.ias.ac.in/fellowship2025/lists/selectedList.jsp

2021

JBNSTS Fellowship – Government of West Bengal

2025

Winner - Abhiprajna 2025 – IISER Tirupati

An all-India team competition of graduate and undergraduate students held annually at IISER Tirupati. It consists of 4 subjects: Mathematics, Physics, Chemistry and Biology with interdisciplinary emphasis.
Scored the highest points out of all participants in the Quiz rounds and presented a prospective area of work on HRV (Heart Rate Variability) modelling using HVGs (Horizontal Visibility Graphs).

Link https://www.abhiprajna.com

PROJECTS

11/2025 – CURRENT

Testing the Efficacy of Horizontal Visibility Graphs (HVGs) in Modeling Heart Rate Variability (HRV)

The idea was introduced as a research topic for presentation at Abhiprajna Competition 2025. A few recent developments on the topic was discovered and were later taken up as a project work to further explore the reliability of HVG as a good model to measure and analyze chaos in HRV. The chaotic pattern of heart rate can be very accurately modelled using time-series analysis of pulse data which can then be analyzed using various preset parameters. This promises to reveal multiple crucial tells on existence of anomalies or diseases in the human heart.
This project is presently under development.

25/05/2025 – CURRENT

A study on calculus of several variables, surfaces and calculus of variations

Part of a study under Prof. K. Sandeep, TIFR-CAM, Bengaluru. This note has been prepared with reference to various online and offline resources, along with the professor's teachings.
I learnt various functional analytical techniques and studies to define familiar notions of several variables. Starting from boundedness, continuity, the Frechet Derivative, chain rule and the mean value theorem, moving on to inverse

and implicit function theorems - I studied surfaces, patches, atlases, tangents and normals from differential geometry using the techniques from before.
All these were then used in the second part of the work: study of calculus of variations.
I learnt multiple topics ranging from the first variation, functionals of higher orders and with multiple independents to the isoperimetric problem, the inverse problem and lagrange multiplier.
A really fascinating part of the study was then to study the previously studied notions of Frechet Derivative and Dirichlet principle using the machinery of calculus of variations.
I am now working on the formulations of these techniques to study Hamiltonians and Noether's Theorem from a purely mathematical standpoint.

Link https://drive.google.com/file/d/1H_n-tgSrItlDKnSbFI4IhxDJog_8YNmc/view?usp=sharing

HOBBIES AND INTERESTS

Chess

- Gold Medalist for IISER Kolkata at IISM 2023
- Top Scorer for IISER Kolkata at Parakram '24
- Avid Chess Fan

Links https://chess-results.com/tnr870679.aspx?lan=1&art=8 | https://chess-results.com/tnr905617.aspx?lan=1

Literature

- 20th Century Bengali Literature
- Story-writer

VOLUNTEERING

2024 – 2025 IISER Kolkata

Secretary - Chess Club, IISER Kolkata

CREATIVE WORKS

2024 – 2025

Lead Actor (2 times) - Annual Drama Production

IISER Kolkata

DIGITAL SKILLS TEST RESULTS

Information and data literacy	**ADVANCED**	Level 6 / 6
Communication and collaboration	**ADVANCED**	Level 6 / 6
Digital content creation	**ADVANCED**	Level 6 / 6
Safety	**ADVANCED**	Level 6 / 6
Problem solving	**ADVANCED**	Level 6 / 6

Results from a self-assessment based on The Digital Competence Framework 2.1